

# **ENV 6128**

## **Smart Air Quality Monitoring & Air Pollution Control**

# Today's Topics

- Introduction to air monitoring network
- Expectation
  - Understand different air monitoring networks in the US

# Air Monitoring Networks

- State or Local Air Monitoring Stations (SLAMS)
- National Core (NCore)
- Photochemical Assessment Monitoring Stations (PAMS)
- PM2.5 Chemical Speciation Network (CSN)
- National Toxics Trends Network (NATTS)
- Interagency Monitoring of Protected Visual Environments (IMPROVE)
- Clean Air Status and Trends Network (CASTNET)
- National Atmospheric Deposition Network (NADP)
- Other non-regulatory or industrial monitors such as Special Purpose Monitors (SPM)
- Tribal air monitoring programs

# State or Local Air Monitoring Stations (SLAMS)

- Biggest network
- Operated by state or local agencies
- Primary purpose: comparison to NAAQS
  - Must use Federal reference method (FRM), Federal equivalent method (FEM), or Approved Regional Method (ARM) monitors
- Additional purposes may also include:
  - Provide air pollution data to the general public in a timely manner
  - Support compliance with air quality standards and emissions strategy development
  - Support air pollution research studies
- SLAMS network include NCore, PAMS, and CSN networks
- Details in CFR Title 40, Parts 50, 53 and 58

# SLAMS location considerations

- The highest concentrations expected to occur in the area covered by the network;
- Typical concentrations in areas of high population density;
- The impact on ambient pollution levels of significant sources or source categories;
- The general background concentration levels;
- The extent of regional pollutant transport among populated areas, and in support of secondary standards; and
- Air pollution impacts on visibility, vegetation damage, or other welfare- based impacts.

# National Core (NCore)

- Part of SLAMS network
- Characterize regional and urban patterns of air pollution
- Comprehensive multi-pollutant measurement
  - PM<sub>2.5</sub>, speciated PM<sub>2.5</sub>, PM<sub>10-2.5</sub>, speciated PM<sub>10-2.5</sub>, O<sub>3</sub>, SO<sub>2</sub>, CO, NO/NO<sub>2</sub>/NO<sub>y</sub>, basic meteorology
- Monitors for all gases (except for O<sub>3</sub>) needs to be more sensitive than standard FRM/FEM monitors
  - Likely some pollutant concentrations are low

# Photochemical Assessment Monitoring Stations (PAMS)

- Part of SLAMS network
- For evaluating, tracking the progress and refining control strategies for ozone control
- Monitor ozone, NO<sub>x</sub>, and VOC to obtain more comprehensive and representative data on ozone air pollution
  - Recall ozone isopleth plot and how ozone control works

# Chemical Speciation Network (CSN)

- Part of SLAMS network
- Provide PM chemical composition data
  - Filter collection → chemical analysis
- Mostly located near larger Metropolitan Statistical Areas, some co-located with PAMS network
- A subset called Speciation Trends Network (STN) provide long-term chemical trend of PM composition
  - Annual and seasonal spatial characterization of aerosols
  - Air quality trends and control progress tracking
  - Compare speciation data with IMPROVE network data
  - Develop emission control strategies

# National Toxics Trends Network (NATTS)

- Main goal: Tracking ambient trends, facilitate tracking progress toward emission and risk reduction goals
- Evaluate public exposure & environmental impacts in the near monitors
- Provide quality assured air toxic data for risk characterization
- Assess effectiveness of specific emission reduction activities
- Evaluate and improve air toxics emission inventories and model performance
- Recall Air toxic screening / NATA

# Interagency Monitoring of Protected Visual Environments (IMPROVE)

- Help protection of visibility in Class I areas
- Goals:
  - Establish current visibility and aerosol conditions in mandatory class I areas
  - Identify chemical species and emission sources responsible for existing visibility impairment
  - Document long-term trends for assessing progress towards the national visibility goal
  - Provide regional haze monitoring representing all visibility-protected class I areas where practical

# Clean Air Status and Trends Network (CASTNET)

- Long-term multipollutant air quality network in rural area
  - 35 years of continuous data
- Main goals
  - Assess the effectiveness of emission reduction programs
  - Support the assessment of the primary and secondary NAAQS
  - Assessing air pollution impacts to sensitive ecosystems
- Many co-locate with National Atmospheric Deposition Program (NADP) monitors

# National Atmospheric Deposition Network (NADP)

- NADP operates several networks
  - National Trends Network (NTN): provides a long-term record of the acids, nutrients, and base cations in U.S. precipitation.
  - Mercury Deposition Network (MDN): provides data on the geographic distributions and trends of mercury in precipitation.
  - Ammonia Monitoring Network (AMoN): measures air concentration of ammonia using passive monitors
  - Atmospheric Mercury Network (AMNet): reports atmospheric mercury for determination of mercury dry deposition
  - Mercury Litterfall Network (MLN): provides an estimate of an important component of mercury dry deposition to a forested landscape